

MedTriage-NLP

Project Write-Up — Build Component

1. Problem

Manually routing patients to the correct hospital department based on their symptom description is slow and inconsistent, leading to delays and potential misrouting. This project builds an NLP text classifier that predicts the most relevant hospital department (e.g., Cardiology, Orthopedics, ENT) directly from a symptom or case description, helping automate the first step of clinical triage.

2. Dataset

The Medical Transcriptions dataset (Kaggle, uploaded by tboyle10, sourced from mtsamples.com) was used, released under a CC0: Public Domain license. The raw dataset contains 4,999 real clinical transcription records spanning 39 medical specialty categories, each with a short description, full transcription text, associated keywords, and a specialty label.

3. Preprocessing

- Removed non-department categories such as "SOAP / Chart / Progress Notes," "Office Notes" and "Consult - History and Phy." — these are document types, not clinical departments.
- Excluded overlapping or non-triage specialties (Surgery, Radiology, Neurosurgery, General Medicine, Oncology) and specialties with too few samples (e.g., Dentistry, Endocrinology). Early testing showed General Medicine and Oncology vocabulary overlapped heavily with other departments, and removing them raised accuracy from 71% to 80%.
- Mapped the remaining labels to 11 hospital departments: Cardiology, Orthopedics, Gastroenterology, Neurology, Gynecology, Urology, ENT, Ophthalmology, Nephrology, Pediatrics and Psychiatry.
- Combined each case's description with its keywords field, after removing any keyword phrase that directly named a department. This adds real clinical signal while avoiding label leakage (using raw keywords would have inflated accuracy to a misleading 91%).
- Final cleaned dataset: 2,232 real cases, split 80/20 (stratified) into 1,855 training and 377 test cases.

4. Approach / Model

Text was vectorized using TF-IDF (unigrams and bigrams, sublinear term frequency scaling, English stop-words removed), producing an 8,000-dimension feature space. A Logistic Regression classifier ($C = 1$, class-balanced weighting) was trained on this representation to predict department labels. Class-balanced weighting prevents the model from favouring larger departments such as Cardiology and Orthopedics. Regularisation strength and vocabulary size were tuned with cross-validation (grid search over C), which together with the keyword features improved test accuracy from 78% to 84.3%.

5. Results

Metric	Score
Overall Accuracy	84.3%
Macro F1-score	0.83
Weighted F1-score	0.85

Performance was strong across most departments (Ophthalmology $F1 = 0.97$, Gynecology $F1 = 0.91$, Cardiology $F1 = 0.90$), with weaker performance on Pediatrics ($F1 = 0.53$) and Neurology ($F1 = 0.69$) due to smaller sample sizes and vocabulary overlap with other departments. In the subgroup audit, accuracy was 82.3% on short descriptions versus 86.6% on long ones, and the top predictive words per department (e.g., chest, cardiac and coronary for Cardiology) matched real clinical vocabulary. The full audit, charts and limitations are in the accompanying one-page Model Card.

6. Tools Used

Python, Pandas, NumPy, Scikit-learn (TfidfVectorizer, LogisticRegression, train_test_split, classification metrics), Matplotlib, Seaborn, and Google Colab as the development environment.